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MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) MGU-BSc(HONOURS) REGULAR EXAMINATION

OCTOBER 2026

FIFTH SEMESTER

Discipline Specific Core Course (DSC) – MG5DSCFTQ300 - FOOD ANALYSIS

(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes **Maximum Marks: 50**

Part A

Short Answer Questions

Answer any 5 questions

Each question carries 2 marks

1. Mention the main steps involved in the procedure of microwave ashing [U]/ [CO1]

Weigh the sample and place it in a suitable microwave digestion vessel.

Add appropriate reagents, heat using the microwave ashing system, cool the vessel, and collect the ash or digest for analysis.

2. Write the disadvantages of Phenol-sulfuric Acid method? [A][CO2]

The method is non-specific and estimates total carbohydrates rather than individual sugars.

Interference may occur due to other sample constituents, and concentrated sulfuric acid requires careful handling.

3. Define oxidation and reduction [K]/ [CO3]

Oxidation: Loss of electrons or an increase in oxidation number.

Reduction: Gain of electrons or a decrease in oxidation number.

4. Mention any two limitations or interfering factors in protein estimation by UV absorption at 280 nm [U] / [CO2]

Nucleic acids and other UV-absorbing substances may interfere with protein estimation.

The method depends on the presence of aromatic amino acids, particularly tryptophan and tyrosine, and may give inaccurate results for proteins with low aromatic amino acid content.

5. What is the purpose of adding oxidizing agents during wet ashing [K]/ [CO1]

To oxidize and destroy organic matter present in the sample.

To convert the sample into a clear solution containing minerals for subsequent analysis.

6. Name any two types of equipment used in the dry ashing method [K]/ [CO1]

Muffle furnace. and Porcelain or silica crucible.

7. Write down the method stated in Englyst-Cummings? [A]/ [CO2]The Englyst–Cummings method is used to determine dietary fibre, particularly non-starch polysaccharides, in foods.It involves enzymatic removal of starch and other digestible components, followed by hydrolysis of non-starch polysaccharides and analysis of the released monosaccharides. [2x5=10]

Part B

Short Essay Questions

Answer any 5 questions

Each question carries 4 marks

8. Discuss the advantages, limitations and precautions of the Gerber method for milk fat determination [U]/ [CO3]

Advantages; It is a rapid, simple and inexpensive method for determining fat content in milk.It is widely used in dairy laboratories for routine milk quality analysis.

Limitations: The method uses concentrated sulphuric acid, which is corrosive and hazardous. Incorrect acid concentration, temperature or centrifugation may lead to inaccurate results.

Precautions: Use the correct quantity and concentration of sulphuric acid and isoamyl alcohol.

Maintain the recommended temperature and centrifugation conditions, and read the fat column at eye level.

9. Describe the colour development reaction in the Somogyi-Nelson method [U]/ [CO2]

Reducing sugars reduce alkaline copper sulphate to cuprous oxide during heating.

Cuprous oxide reduces arsenomolybdate reagent to form a blue-coloured complex, whose intensity is measured spectrophotometrically and is proportional to the reducing sugar concentration.

10. Explain the principle of Vitamin C estimation by the DCPIP method [U]/ [CO3]

Vitamin C (ascorbic acid) reduces the blue-coloured 2,6-dichlorophenolindophenol (DCPIP) dye to its colourless reduced form.The amount of DCPIP decolourized is proportional to the vitamin C content of the sample. The endpoint is indicated by a persistent light pink colour when a slight excess of dye remains.

11. Describe the construction, working principle and application of a hydrometer in food analysis [U]/ [CO1]

Construction: A hydrometer consists of a sealed glass tube with a weighted bulb at the bottom and a graduated stem at the top.

Working principle:. It works on Archimedes' principle: the hydrometer floats at a level determined by the density or specific gravity of the liquid.

Applications: Measuring the specific gravity of milk, fruit juices, syrups and sugar solutions.

Estimating the concentration of dissolved substances in food liquids using suitable calibrated hydrometers.

12. Describe the different steps involved in the Kjeidahl method: digestion, distillation and titration. [U]/ [CO2]

Digestion: The sample is heated with concentrated sulphuric acid and a suitable catalyst, converting organic nitrogen into ammonium sulphate.

Distillation: The digest is made alkaline with sodium hydroxide, releasing ammonia, which is distilled and collected in boric acid or a suitable receiving solution.

Titration: The collected ammonia is titrated with a standard acid solution to determine the nitrogen content.

Calculation: The nitrogen percentage is calculated and multiplied by an appropriate conversion factor to estimate crude protein content.

13. Differentiate between Mohr's method and Volhard's method [U]/[CO3]

Mohr's method	Volhard's method
1. It is a direct argentometric titration method.	1. It is generally a back-titration method for chloride estimation.

2. Potassium chromate is used as the indicator.	2. Ferric alum is used as the indicator.
3. The endpoint is indicated by the formation of a brick-red silver chromate precipitate.	3. The endpoint is indicated by the appearance of a reddish-brown ferric thiocyanate complex.
4. It is performed in a nearly neutral medium.	4. It is performed in an acidic medium, usually with nitric acid.

14. Explain the principle behind Karl-Fischer titration and add a note on KFR eq [An]/[CO1] [4x5=20]

1. Principle: Karl Fischer titration is based on the quantitative reaction of water with iodine and sulphur dioxide in the presence of an alcohol and a base. The amount of iodine consumed is proportional to the water content of the sample.

2. KFR equivalent (Karl Fischer Reagent equivalent): It is the amount of water, usually expressed in milligrams, that reacts with 1 mL of Karl Fischer reagent. It is determined by standardization and is used to calculate the moisture content of the sample.

Part C

Long Essay Question
Answer any 2 questions

15. Differentiate between crude fibre and dietary fiber. Explain their importance in food analysis [An]/[CO2]

Crude fibre is the residue remaining after acid and alkali digestion, whereas dietary fibre includes non-digestible food components such as cellulose, hemicellulose, lignin, pectin and resistant starch.

Dietary fibre analysis provides a more accurate assessment of fibre content for nutritional labelling and health evaluation, while crude fibre determination is mainly used in conventional food composition analysis.

16. Compare simple distillation and reflux distillation with an immiscible solvent. Discuss their principles, procedures, advantages, limitations and applications in food analysis with suitable examples. [U]/[CO1]

1. **Simple distillation** separates volatile components based on differences in boiling points by heating the sample, condensing the vapour, and collecting the distillate; it is simple, rapid, and useful for recovering solvents or volatile compounds in food analysis, but may be ineffective for close-boiling components and can cause thermal degradation.2. **Reflux distillation** with an immiscible solvent (azeotropic distillation) involves boiling the food sample with a solvent such as toluene or xylene, condensing the vapours, and separating water from the solvent in a calibrated trap; it is useful for determining moisture in foods such as spices and oils, but requires specialized apparatus, flammable solvents, and careful handling.

17.Evaluate the importance of determining soluble, insoluble and total dietary fiber in food and nutritional analysis [E]/[CO2]

Soluble fibre helps regulate blood glucose and cholesterol, while insoluble fibre increases stool bulk and supports bowel regularity. Total dietary fibre provides an overall measure of fibre content for nutritional labelling, quality control, dietary assessment and the development of fibre-enriched foods.

18. Write a detailed essay on the Modified Mojonnier method as a discontinuous solvent extraction technique for milk fat estimation. Include its principle, characteristics, step by step procedure, precautions, advantages, limitations and applications in dairy and food analysis. [A]/[CO3]

Principle and procedure: The Modified Mojonnier method is a discontinuous solvent extraction technique. Ammonia helps release fat from the protein matrix, ethanol assists extraction, and diethyl ether and petroleum ether extract the fat. The organic layers are separated, combined, and evaporated; the remaining fat is dried and weighed to calculate the fat percentage. **Precautions, advantages, limitations and applications**: Use clean, dry glassware, ensure complete extraction and phase separation, avoid overheating, and handle flammable solvents in a fume hood away from ignition sources. The method provides reliable gravimetric fat estimation for dairy quality control and nutritional labelling, but it is time-consuming, requires skilled handling and may produce errors from incomplete extraction or solvent transfer losses.

[10x2=20]